

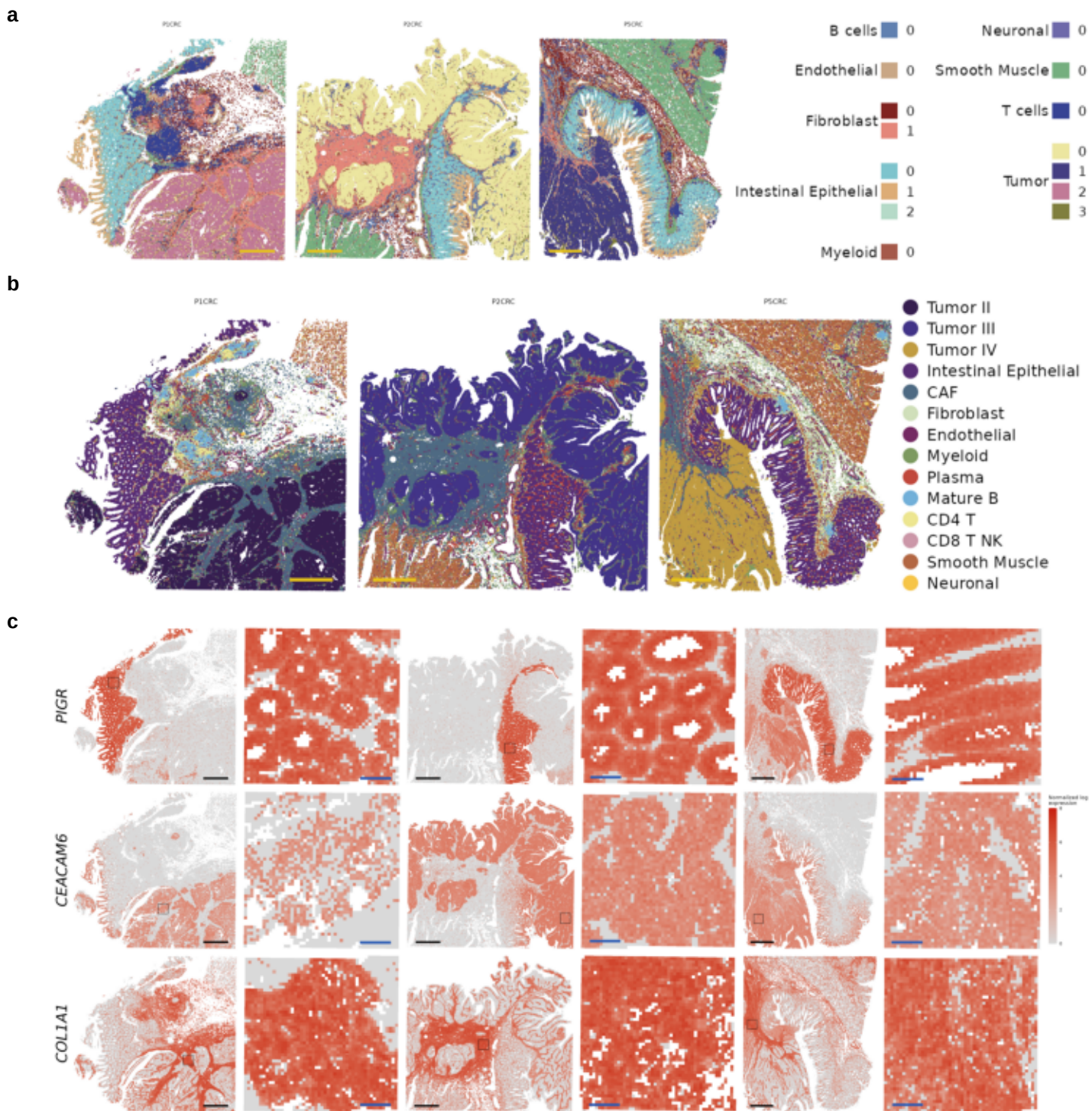
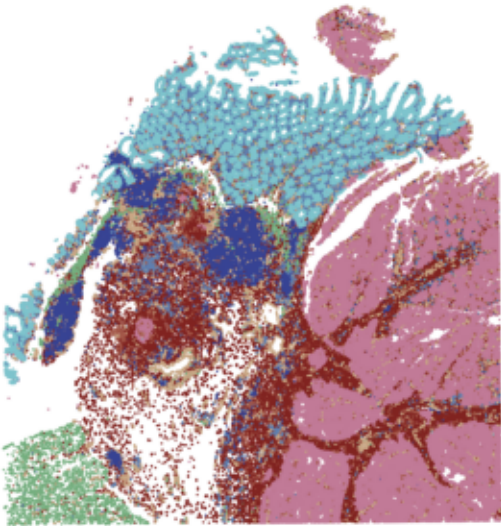


Fig. 3 | Spatial mapping of CRC samples using Visium HD (reproduction).

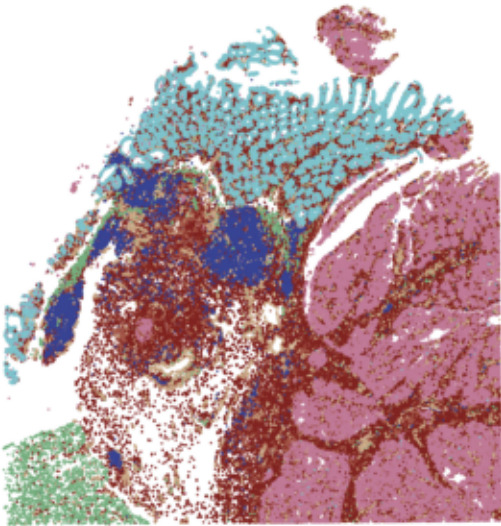
a, Three CRC samples (P1CRC, P2CRC and P5CRC), 8-um bins colored by unsupervised cluster from a joint analysis of all samples. b, The same samples, bins colored by deconvolution cell type (14-class single-cell reference). CAF, cancer-associated fibroblast. c, PIGR (goblet cells

and enterocytes), CEACAM6 (tumor) and COL1A1 (fibroblasts) expression; tissue view on the left, inset (black box) magnified on the right. Scale bars, 1 mm (yellow or black), 80 um (blue).

P1CRC — R (Seurat)

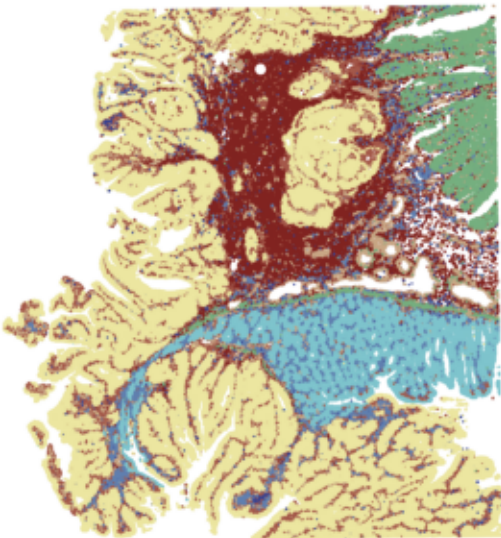


P1CRC — Python (scanpy)

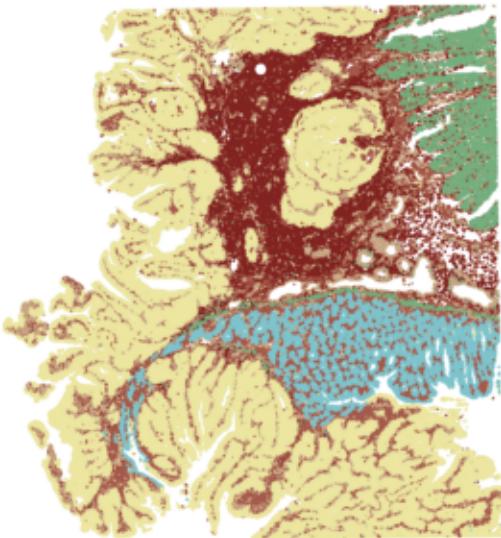


Clustering concordance — R vs. Oliveira et al. (2025): ARI = 0.816 | R vs. Python: ARI = 0.454

P2CRC — R (Seurat)

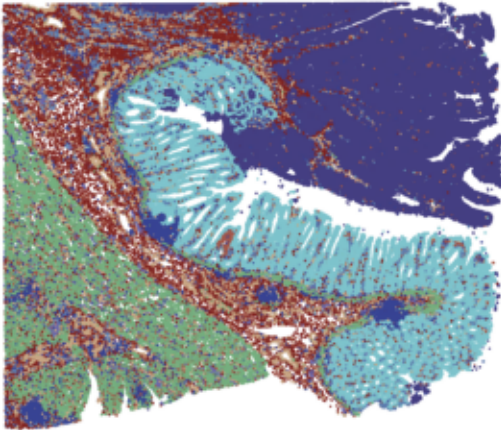


P2CRC — Python (scanpy)

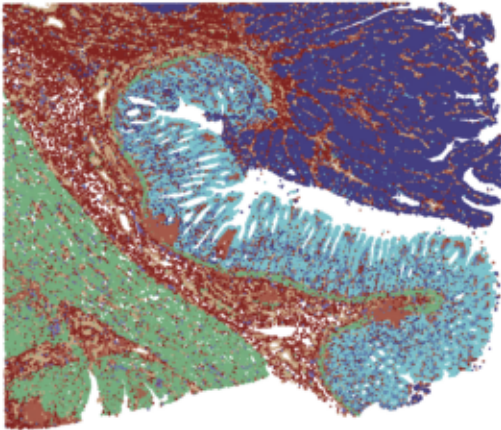


Clustering concordance — R vs. Oliveira et al. (2025): ARI = 0.605 | R vs. Python: ARI = 0.418

P5CRC — R (Seurat)



P5CRC — Python (scanpy)



Clustering concordance — R vs. Oliveira et al. (2025): ARI = 0.730 | R vs. Python: ARI = 0.498

Extended Data Fig. 1 | R vs. Python clustering.
Joint clustering computed independently in R (Seurat, Louvain) and Python (scanpy, Leiden), shown per patient.

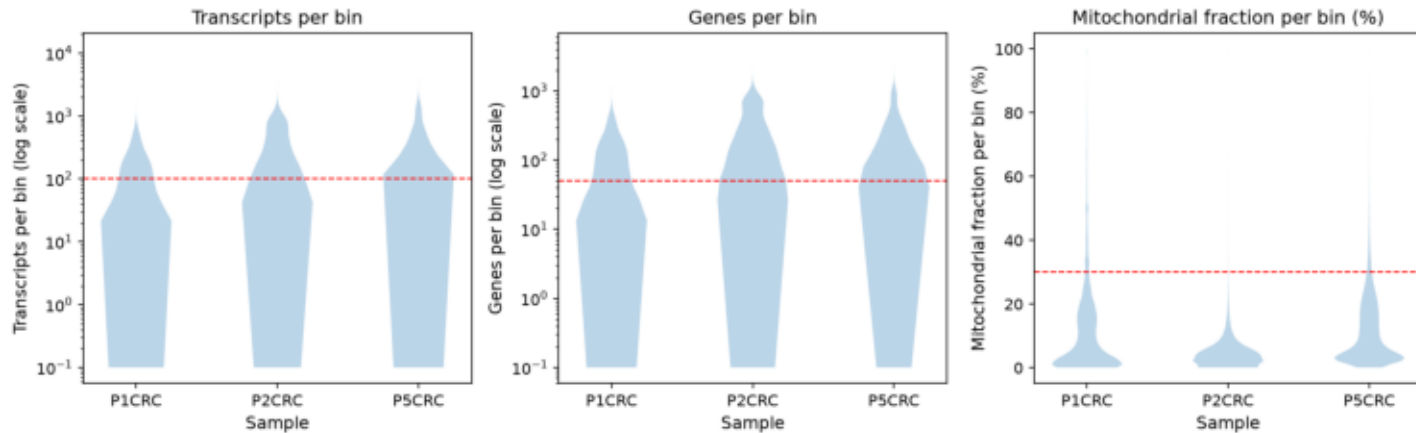
ARI on shared bins is given below each pair: R vs. Oliveira et al. (2025), and R vs. Python.

Table 1 | Deconvolution benchmark

Sample	Agreement
P1CRC	91.7%
P2CRC	91.8%
P5CRC	91.4%

Per-bin agreement between RCTD cell types (14-class reference) and the deconvolution labels of Oliveira et al. (2025). Bins matched by barcode; rejected bins excluded.

QC per 8 um bin — violin width: number of bins; dashed line: filter threshold (≥ 100 transcripts, ≥ 50 genes, $< 30\%$ mito)



Extended Data Fig. 2 | Bin-level QC.

Transcripts, genes and mitochondrial fraction per 8-um bin. Dashed lines, filter thresholds (≥ 100 transcripts,

≥ 50 genes, $< 30\%$ mito). Bins retained: 286,973 (P1CRC), 428,405 (P2CRC), 373,156 (P5CRC).